



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of CSE (Artificial Intelligence and Machine Learning)

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: I Semester – B.E. CSE (AIML)

Course Code & Title: CS25C01 Computer Programming: C

Name of the Faculty member (s): Dr. R. Venkatesh, ASP & Head /CSE (AIML)

Date of Implementation : 16.09.2025

Innovative Practice Description

- **Unit / Topic: Module I - Logical, Bitwise, Assignment, Conditional operators**
- **Course Outcome: CO1**
- **Activity Chosen: Flash Cards**
- **Justification:**
 - Flash cards are a memory aid tool that helps in quick recall and active learning.
 - Logical, Bitwise, Assignment, and Conditional operators form the foundation of decision-making and expression evaluation in C programming.
 - To enhance conceptual clarity and to improve recall of operators, their symbols, and typical outputs, the flash card activity was chosen.
 - Quizlet, an online flash-card platform, allows students to learn through multiple modes such as Flashcards, Match, Learn, Test, and Live Team Mode, making the activity more interactive and effective.
- **Time Allotted for the Activity: 15 Minutes**
- **Pre Implementation:**
 - Two sets of flash cards were created using Quizlet (<https://quizlet.com/>):
 - One set with questions related to
 - logical operators and their usage,
 - bitwise operators and their outputs,
 - compound assignment operators,
 - conditional (ternary) operator.
 - Another set with answers, including
 - definitions,
 - operator symbols,
 - short code snippets,
 - expected output values.
- Question and answer cards were organized as separate items inside the Quizlet set.
- Students were instructed to join the Quizlet set using a shared classroom link.
- All student register number were added to the wheel.
- The faculty spun the wheel to randomly select the next student who would read the card, match the card, and explain the concept.

- **Details of the Implementation:**

- The activity was conducted using Quizlet flash cards, and students accessed the flash card set through a shared link provided by the faculty.
- The faculty used a digital Spin-the-Wheel tool (such as Wheel of Names) to randomly select students during the activity.
- When the wheel selected a student, the student read aloud the definition, operator explanation, or output displayed on their Quizlet flash card.
- The wheel was spun again to randomly select another student, who identified the corresponding matching question from the Quizlet set.
- After identifying the correct match, the selected student explained both the question and its correct answer to the entire class using the Quizlet flash card displayed on screen.
- This approach ensured full-class involvement, as any student could be selected at any moment, maintaining attention and interest.
- Students then practiced individually using Quizlet's Match Mode and Learn Mode, reinforcing their understanding of Logical, Bitwise, Assignment, and Conditional operators.
- The Figure 1 shows the sample activity of quizlet.com.
- Students such as Hiruthikraj P and Shereen Salwa U R actively participated when their names were selected by the spin wheel, as shown in Figure 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	Students understand operator functionalities.	Students choose correct operators in problem-solving.	Students apply operators in C expressions.	Collaborative Quizlet activity enhances teamwork.	Students improve communication while explaining answers.	Analytical skills improve through operator-based matching tasks.

• **Images / Screenshot of the practice:**

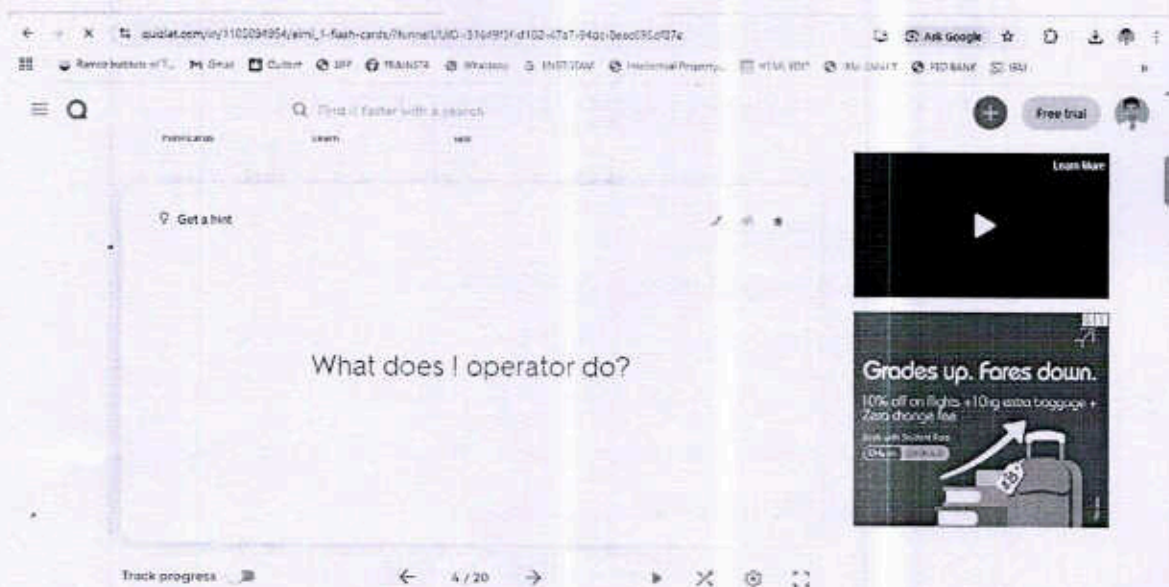


Fig 1: Sample of Flash cards

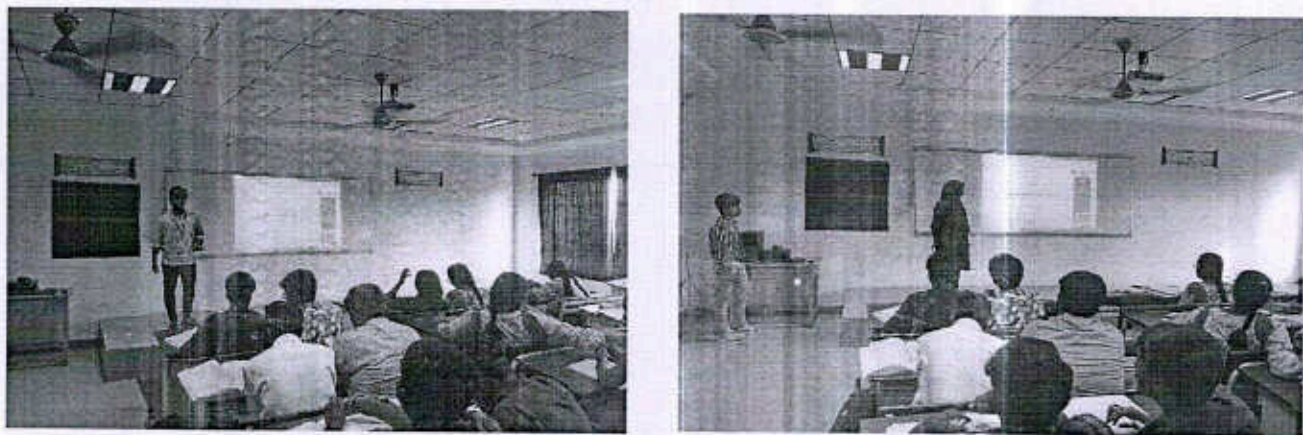


Fig 2: Students (Hiruthikraj P and Shereen Salwa U R) involved in Flash Card Activity

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students found the Quizlet-based flash card activity highly engaging and modern.
- They appreciated the "Match" and "Learn" modes of Quizlet for self-practice.
- Students requested more Quizlet-based activities for other programming concepts.

❖ **Benefit of the practice:**

- Improved recall of operator symbols and meanings.
- Better understanding of logical and bitwise operations through repeated Quizlet practice.
- Increased student engagement and active learning using digital tools.

❖ **Challenges faced in implementation:**

- Some students needed assistance in accessing Quizlet or switching between modes.
- Bitwise operator outputs required slower explanation for beginners.

References:

- <https://quizlet.com/>
- <https://en.wikipedia.org/wiki/Flashcard>
- <https://ellii.com/blog/10-ways-to-use-flashcards-in-class>
- <https://www.geeksforgeeks.org/operators-in-c/>
- https://www.tutorialspoint.com/cprogramming/c_operators.htm
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- <https://www.teachingenglish.org.uk/article/using-flash-cards-young-learners>
- <https://mailchimp.com/resources/what-is-email-automation/>
- <https://www.hubspot.com/email-marketing-automation>
- <https://www.salesforce.com/products/marketing-cloud/email-marketing/automation/>



Signature of Faculty Member



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